Practical No. 8 : Intelligent Agent Guessing Number

```
import random
class NumberGuessingAgent:
    def __init__(self, low=1, high=100):
        self.low = low
        self.high = high

    def guess(self):
        return (self.low + self.high) // 2

    def update(self, guess_value, feedback):
        if feedback == "lower":
            self.high = guess_value - 1
        elif feedback == "higher":
            self.low = guess_value + 1

    def play(self, secret_number):
        attempts = 0

        while self.low <= self.high:
            attempts += 1
            current_guess = self.guess()

            print(f"Attempt {attempts}: Agent guesses {current_guess}")

            if current_guess == secret_number:
                print("Feedback: Correct")
                print(
                    f"Agent guessed the number {secret_number} "
                    f"correctly in {attempts} attempts!"
                )
```

```
            return attempts

        elif current_guess < secret_number:
            print("Feedback: Higher")
            self.update(current_guess, "higher")

        else:
            print("Feedback: Lower")
            self.update(current_guess, "lower")

    print("Range exhausted.")
    return attempts

if __name__ == "__main__":
    LOW, HIGH = 1, 100
    secret_number = random.randint(LOW, HIGH)
    agent = NumberGuessingAgent(LOW, HIGH)
    agent.play(secret_number)
```